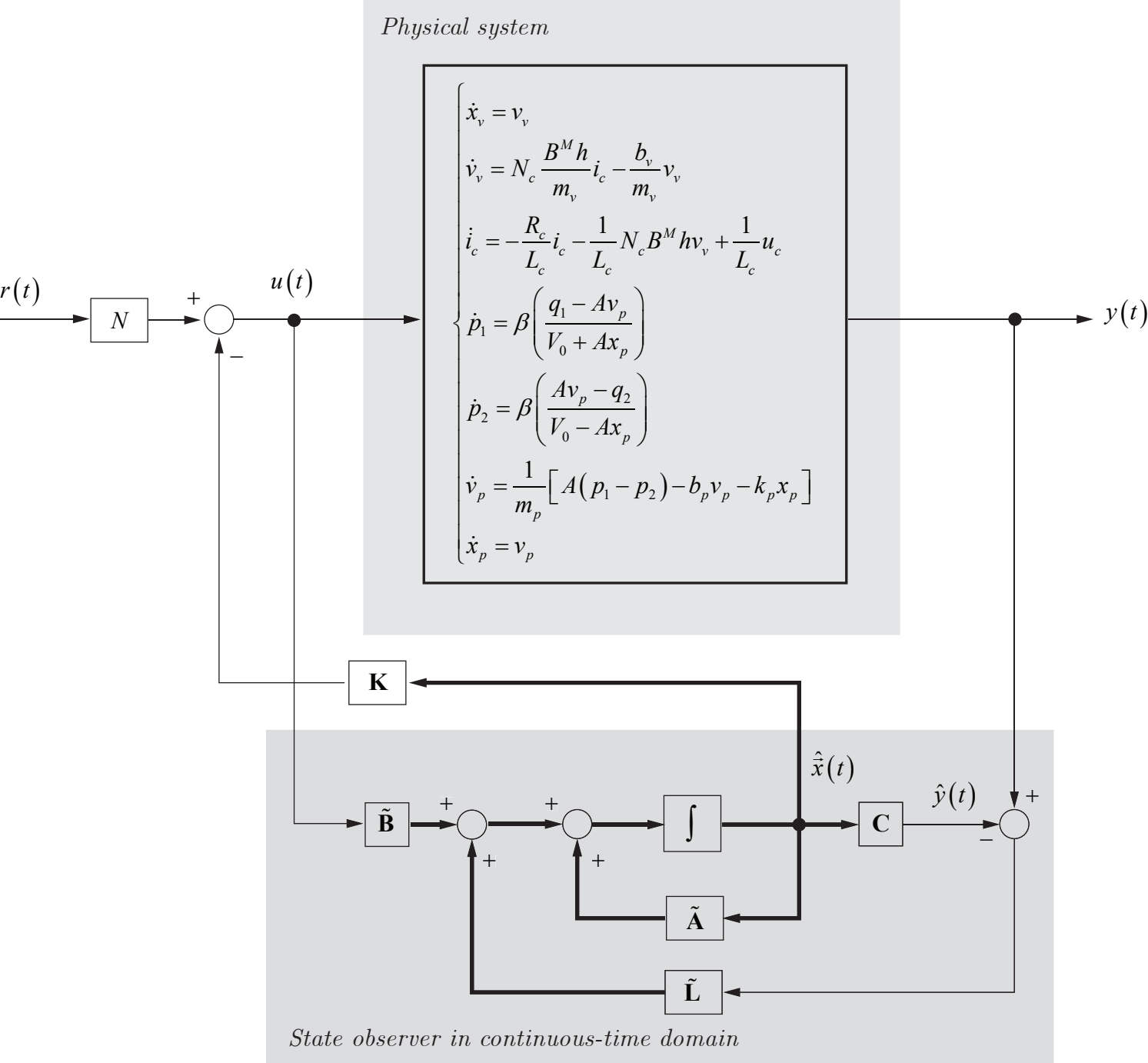

Physical system
$\dot{x}_v = v_v$
$\dot{v}_v = N_c \frac{B^M h}{m_v} i_c - \frac{b_v}{m_v} v_v$
$\dot{i}_c = -\frac{R_c}{L_c} i_c - \frac{1}{L_c} N_c B^M h v_v + \frac{1}{L_c} u_c$
$\dot{p}_1 = \beta \left(\frac{q_1 - A v_p}{V_0 + A x_p} \right)$
$\dot{p}_2 = \beta \left(\frac{A v_p - q_2}{V_0 - A x_p} \right)$
$\dot{v}_p = \frac{1}{m_p} \left[A(p_1 - p_2) - b_p v_p - k_p x_p \right]$
$\dot{x}_p = v_p$
$r(t)$
N
$u(t)$
$y(t)$
K
$\hat{\dot{x}}(t)$
$\hat{y}(t)$
$\tilde{\mathbf{B}}$
$\int$
$\mathbf{C}$
$\tilde{\mathbf{A}}$
$\tilde{\mathbf{L}}$
State observer in continuous-time domain